13–(MathAA/10_HL_Winter_2022_Q10)–*Differentiation,Graphs*

[Maximum mark: 20]

The function f is defined by $f(x) = \cos^2 x - 3\sin^2 x$, $0 \le x \le \pi$.

(a) Find the roots of the equation $f(x) = 0$. [5]

(b) (i) Find $f'(x)$.

(ii) Hence find the coordinates of the points on the graph of $y = f(x)$ where $f'(x) = 0$. [7]

(c) Sketch the graph of $y = |f(x)|$, clearly showing the coordinates of any points where $f'(x) = 0$ and any points where the graph meets the coordinate axes. [4]

(d) Hence or otherwise, solve the inequality $|f(x)| > 1$. [4]

1-(MAT/11_HL_Summer_2012_Q3)-*ComplexNumbers*

[Maximum mark: 7]

If $z_1 = a + a\sqrt{3}\,\mathrm{i}$ and $z_2 = 1 - \mathrm{i}$, where a is a real constant, express z_1 and z_2 in the form $r\,\mathrm{cis}\,\theta$, and hence find an expression for $\left(\frac{z_1}{z_2}\right)^6$ in terms of a and i.

2-(MAT/12_HL_Summer_2012_Q6)**-***ComplexNumbers*

[Maximum mark: 7]

Given that $(4-5\mathrm{i})m+4n=16+15\mathrm{i}$, where $\mathrm{i}^2=-1$, find m and n if

(a) m and n are real numbers; *[3 marks]*

(b) m and n are conjugate complex numbers. *[4 marks]*

3-(MAT/11_HL_Summer_2012_Q7)-*ComplexNumbers*

[Maximum mark: 6]

Given that z is the complex number $x+\mathrm{i}y$ and that $|z|+z=6-2\mathrm{i}$, find the value of x and the value of y.

4–(MAT/12_HL_Summer_2012_Q12)**–***ComplexNumbers*

Part A *[Maximum mark: 12]*

(a) Given that $(x+\mathrm{i}y)^2 = -5+12\mathrm{i}$, x, $y \in \mathbb{R}$. Show that

(i) $x^2 - y^2 = -5$;

(ii) $xy = 6$. *[2 marks]*

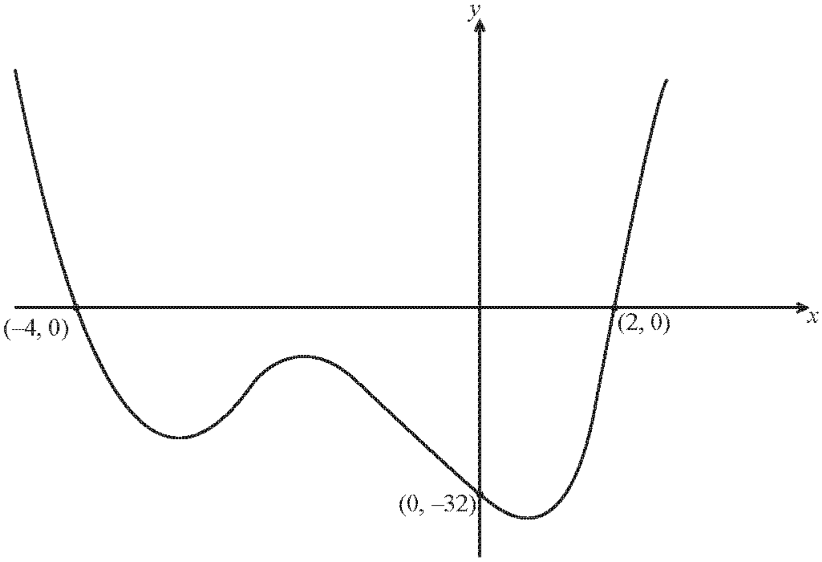
(b) Hence find the two square roots of $-5+12\mathrm{i}$. *[5 marks]*

(c) For any complex number z, show that $(z^*)^2 = (z^2)^*$. *[3 marks]*

(d) Hence write down the two square roots of $-5-12\mathrm{i}$. *[2 marks]*

Part B *[Maximum mark: 17]*

The graph of a polynomial function f of degree 4 is shown below.



(a) Explain why, of the four roots of the equation $f(x)=0$, two are real and two are complex. *[2 marks]*

(b) The curve passes through the point $(-1, -18)$. Find $f(x)$ in the form $f(x) = (x-a)(x-b)(x^2+cx+d)$, where $a,\ b,\ c,\ d \in \mathbb{Z}$. *[5 marks]*

(c) Find the two complex roots of the equation $f(x) = 0$ in Cartesian form. *[2 marks]*

(d) Draw the four roots on the complex plane (the Argand diagram). *[2 marks]*

(e) Express each of the four roots of the equation in the form $re^{i\theta}$. *[6 marks]*

5-(MAT/10_HL_Winter_2012_Q10)**-***ComplexNumbers*

Consider the complex numbers

$$z_1 = 2\sqrt{3}\,\text{cis}\,\frac{3\pi}{2} \text{ and } z_2 = -1+\sqrt{3}\text{i}\,.$$

(a) (i) Write down z_1 in Cartesian form.

(ii) Hence determine $(z_1+z_2)^*$ in Cartesian form. *[3 marks]*

(b) (i) Write z_2 in modulus-argument form.

(ii) Hence solve the equation $z^3 = z_2$. *[6 marks]*

(c) Let $z = r\,\text{cis}\,\theta$, where $r \in \mathbb{R}^+$ and $0 \le \theta < 2\pi$. Find all possible values of r and θ,

(i) if $z^2 = (1+z_2)^2$;

(ii) if $z = -\dfrac{1}{z_2}$. *[6 marks]*

(d) Find the smallest positive value of n for which $\left(\dfrac{z_1}{z_2}\right)^n \in \mathbb{R}^+$. *[4 marks]*

6-(MAT/11_HL_Summer_2013_Q1)-*ComplexNumbers*

(a) If $w = 2 + 2\mathrm{i}$, find the modulus and argument of w. *[2 marks]*

(b) Given $z = \cos\left(\frac{5\pi}{6}\right) + \mathrm{i}\sin\left(\frac{5\pi}{6}\right)$, find in its simplest form $w^4 z^6$. *[4 marks]*

7-(MAT/12_HL_Summer_2013_Q7)-*ComplexNumbers*

Given the complex numbers $z_1 = 1 + 3\mathrm{i}$ and $z_2 = -1 - \mathrm{i}$.

(a) Write down the exact values of $|z_1|$ and $\arg(z_2)$. *[2 marks]*

(b) Find the minimum value of $|z_1 + \alpha z_2|$, where $\alpha \in \mathbb{R}$. *[5 marks]*

8–(MAT/12_HL_Summer_2013_Q13)–*ComplexNumbers*

(a) (i) Express each of the complex numbers $z_1 = \sqrt{3} + \mathrm{i}$, $z_2 = -\sqrt{3} + \mathrm{i}$ and $z_3 = -2\mathrm{i}$ in modulus-argument form.

(ii) Hence show that the points in the complex plane representing z_1, z_2 and z_3 form the vertices of an equilateral triangle.

(iii) Show that $z_1^{3n} + z_2^{3n} = 2z_3^{3n}$ where $n \in \mathbb{N}$. *[9 marks]*

(b) (i) State the solutions of the equation $z^7 = 1$ for $z \in \mathbb{C}$, giving them in modulus-argument form.

(ii) If w is the solution to $z^7 = 1$ with least positive argument, determine the argument of $1 + w$. Express your answer in terms of π.

(iii) Show that $z^2 - 2z\cos\left(\frac{2\pi}{7}\right) + 1$ is a factor of the polynomial $z^7 - 1$. State the two other quadratic factors with real coefficients. *[9 marks]*

9–(MAT/10_HL_Winter_2013_Q12)–*ComplexNumbers,Trigonometry,Integration*

Consider the complex number $z = \cos\theta + \mathrm{i}\sin\theta$.

(a) Use De Moivre's theorem to show that $z^n + z^{-n} = 2\cos n\theta$, $n \in \mathbb{Z}^+$. *[2]*

(b) Expand $\left(z + z^{-1}\right)^4$. *[1]*

(c) Hence show that $\cos^4\theta = p\cos 4\theta + q\cos 2\theta + r$, where p, q and r are constants to be determined. *[4]*

(d) Show that $\cos^6\theta = \dfrac{1}{32}\cos 6\theta + \dfrac{3}{16}\cos 4\theta + \dfrac{15}{32}\cos 2\theta + \dfrac{5}{16}$. *[3]*

(e) Hence find the value of $\int_0^{\frac{\pi}{2}} \cos^6\theta \, \mathrm{d}\theta$. *[3]*

The region S is bounded by the curve $y = \sin x \cos^2 x$ and the x-axis between $x = 0$ and $x = \dfrac{\pi}{2}$.

(f) S is rotated through 2π radians about the x-axis. Find the value of the volume generated. *[4]*

(g) (i) Write down an expression for the constant term in the expansion of $\left(z + z^{-1}\right)^{2k}$, $k \in \mathbb{Z}^+$.

(ii) Hence determine an expression for $\int_0^{\frac{\pi}{2}} \cos^{2k}\theta \, \mathrm{d}\theta$ in terms of k. *[3]*

10–(MAT/12_HL_Summer_2014_Q7)–*ComplexNumbers*

Consider the complex numbers $u = 2 + 3\mathrm{i}$ and $v = 3 + 2\mathrm{i}$.

(a) Given that $\frac{1}{u} + \frac{1}{v} = \frac{10}{w}$, express w in the form $a + b\mathrm{i}$, $a, b \in \mathbb{R}$. *[4]*

(b) Find w^* and express it in the form $re^{\mathrm{i}\theta}$. *[3]*

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11–(MAT/11_HL_Summer_2014_Q13)–*ComplexNumbers,Sequences-Series*

A geometric sequence $\{u_n\}$, with complex terms, is defined by $u_{n+1} = (1+\mathrm{i})u_n$ and $u_1 = 3$.

(a) Find the fourth term of the sequence, giving your answer in the form $x + y\mathrm{i}$, $x,\ y \in \mathbb{R}$. *[3]*

(b) Find the sum of the first 20 terms of $\{u_n\}$, giving your answer in the form $a \times (1 + 2^m)$ where $a \in \mathbb{C}$ and $m \in \mathbb{Z}$ are to be determined. *[4]*

A second sequence $\{v_n\}$ is defined by $v_n = u_n u_{n+k}$, $k \in \mathbb{N}$.

(c) (i) Show that $\{v_n\}$ is a geometric sequence.

(ii) State the first term.

(iii) Show that the common ratio is independent of k. *[5]*

A third sequence $\{w_n\}$ is defined by $w_n = |u_n - u_{n+1}|$.

(d) (i) Show that $\{w_n\}$ is a geometric sequence.

(ii) State the geometrical significance of this result with reference to points on the complex plane. *[5]*

12–(MAT/10_HL_Winter_2014_Q13)–*ComplexNumbers,Trigonometry,Integration*

(a) (i) Show that $(1+\mathrm{i}\tan\theta)^n+(1-\mathrm{i}\tan\theta)^n=\dfrac{2\cos n\theta}{\cos^n\theta}$, $\cos\theta\neq 0$.

(ii) Hence verify that $\mathrm{i}\tan\dfrac{3\pi}{8}$ is a root of the equation $(1+z)^4+(1-z)^4=0$, $z\in\mathbb{C}$.

(iii) State another root of the equation $(1+z)^4+(1-z)^4=0$, $z\in\mathbb{C}$. *[10]*

(b) (i) Use the double angle identity $\tan 2\theta=\dfrac{2\tan\theta}{1-\tan^2\theta}$ to show that $\tan\dfrac{\pi}{8}=\sqrt{2}-1$.

(ii) Show that $\cos 4x=8\cos^4 x-8\cos^2 x+1$.

(iii) Hence find the value of $\displaystyle\int_0^{\frac{\pi}{8}}\frac{2\cos 4x}{\cos^2 x}\,\mathrm{d}x$. *[13]*

13–(MAT/12_HL_Summer_2015_Q7)–*ComplexNumbers*

(a) Find three distinct roots of the equation $8z^3 + 27 = 0$, $z \in \mathbb{C}$ giving your answers in modulus-argument form. [6]

The roots are represented by the vertices of a triangle in an Argand diagram.

(b) Show that the area of the triangle is $\dfrac{27\sqrt{3}}{16}$. [3]

14–(MAT/10_HL_Winter_2015_Q11)–*ComplexNumbers*

(a) Solve the equation $z^3 = 8\mathrm{i}$, $z \in \mathbb{C}$ giving your answers in the form $z = r(\cos\theta + \mathrm{i}\sin\theta)$ and in the form $z = a + b\mathrm{i}$ where $a, b \in \mathbb{R}$. [6]

(b) Consider the complex numbers $z_1 = 1 + \mathrm{i}$ and $z_2 = 2\left(\cos\left(\frac{\pi}{6}\right) + \mathrm{i}\sin\left(\frac{\pi}{6}\right)\right)$.

(i) Write z_1 in the form $r(\cos\theta + \mathrm{i}\sin\theta)$.

(ii) Calculate $z_1 z_2$ and write in the form $z = a + b\mathrm{i}$ where $a, b \in \mathbb{R}$.

(iii) Hence find the value of $\tan\frac{5\pi}{12}$ in the form $c + d\sqrt{3}$, where $c, d \in \mathbb{Z}$.

(iv) Find the smallest value $p > 0$ such that $(z_2)^p$ is a positive real number. [11]

15-(MAT/11_HL_Summer_2016_Q12)-*ComplexNumbers*

(a) Use de Moivre's theorem to find the value of $\left(\cos\left(\frac{\pi}{3}\right)+\mathrm{i}\sin\left(\frac{\pi}{3}\right)\right)^3$. [2]

(b) Use mathematical induction to prove that

$$(\cos\theta-\mathrm{i}\sin\theta)^n=\cos n\theta-\mathrm{i}\sin n\theta \text{ for } n\in\mathbb{Z}^+.$$ [6]

Let $z=\cos\theta+\mathrm{i}\sin\theta$.

(c) Find an expression in terms of θ for $(z)^n+(z^*)^n$, $n\in\mathbb{Z}^+$ where z^* is the complex conjugate of z. [2]

(d) (i) Show that $zz^*=1$.

(ii) Write down the binomial expansion of $(z+z^*)^3$ in terms of z and z^*.

(iii) Hence show that $\cos 3\theta=4\cos^3\theta-3\cos\theta$. [5]

(e) Hence solve $4\cos^3\theta-2\cos^2\theta-3\cos\theta+1=0$ for $0\le\theta<\pi$. [6]

16–(MAT/12_HL_Summer_2016_Q12)–*ComplexNumbers*

Let $w = \cos\frac{2\pi}{7} + \mathrm{i}\sin\frac{2\pi}{7}$.

(a) Verify that w is a root of the equation $z^7 - 1 = 0,\ z \in \mathbb{C}$. [3]

(b) (i) Expand $(w-1)\left(1 + w + w^2 + w^3 + w^4 + w^5 + w^6\right)$.

(ii) Hence deduce that $1 + w + w^2 + w^3 + w^4 + w^5 + w^6 = 0$. [3]

(c) Write down the roots of the equation $z^7 - 1 = 0,\ z \in \mathbb{C}$ in terms of w and plot these roots on an Argand diagram. [3]

Consider the quadratic equation $z^2 + bz + c = 0$ where $b, c \in \mathbb{R},\ z \in \mathbb{C}$. The roots of this equation are α and α^* where α^* is the complex conjugate of α.

(d) (i) Given that $\alpha = w + w^2 + w^4$, show that $\alpha^* = w^6 + w^5 + w^3$.

(ii) Find the value of b and the value of c. [10]

(e) Using the values for b and c obtained in part (d)(ii), find the imaginary part of α, giving your answer in surd form. [4]

17-(MAT/10_HL_Winter_2016_Q12)-*ComplexNumbers*

Let ω be one of the non-real solutions of the equation $z^3 = 1$.

(a) Determine the value of

(i) $1 + \omega + \omega^2$;

(ii) $1 + \omega^* + (\omega^*)^2$. [4]

(b) Show that $(\omega - 3\omega^2)(\omega^2 - 3\omega) = 13$. [4]

Consider the complex numbers $p = 1 - 3\mathrm{i}$ and $q = x + (2x + 1)\mathrm{i}$, where $x \in \mathbb{R}$.

(c) Find the values of x that satisfy the equation $|p| = |q|$. [5]

(d) Solve the inequality $\mathrm{Re}(pq) + 8 < (\mathrm{Im}(pq))^2$. [6]

18-(MAT/11_HL_Summer_2017_Q2)-*ComplexNumbers*

Consider the complex numbers $z_1 = 1 + \sqrt{3}\,\mathrm{i}$, $z_2 = 1 + \mathrm{i}$ and $w = \frac{z_1}{z_2}$.

(a) By expressing z_1 and z_2 in modulus-argument form write down

(i) the modulus of w;

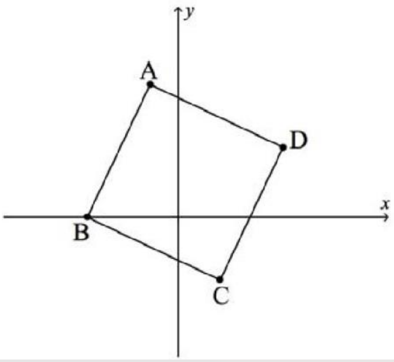
(ii) the argument of w. [4]

(b) Find the smallest positive integer value of n, such that w^n is a real number. [2]

19-(MAT/12_HL_Summer_2017_Q5)-*ComplexNumbers*

[Maximum mark: 4]

In the following Argand diagram the point A represents the complex number $-1+4\mathrm{i}$ and the point B represents the complex number $-3+0\mathrm{i}$. The shape of ABCD is a square. Determine the complex numbers represented by the points C and D.



20–(MAT/12_HL_Summer_2017_Q11)–*Trigonometry,ComplexNumbers*

(a) Solve $2\sin(x+60^\circ)=\cos(x+30^\circ)$, $0^\circ \le x \le 180^\circ$. [5]

(b) Show that $\sin 105^\circ + \cos 105^\circ = \dfrac{1}{\sqrt{2}}$. [3]

(c) Let $z = 1 - \cos 2\theta - \mathrm{i}\sin 2\theta$, $z \in \mathbb{C}$, $0 \le \theta \le \pi$.

(i) Find the modulus and argument of z in terms of θ. Express each answer in its simplest form.

(ii) Hence find the cube roots of z in modulus-argument form. [14]

21-(MAT/10_HL_Winter_2017_Q8)-*ComplexNumbers*

[Maximum mark: 7]

Determine the roots of the equation $(z+2\mathrm{i})^3 = 216\mathrm{i}$, $z \in \mathbb{C}$, giving the answers in the form $z = a\sqrt{3} + b\mathrm{i}$ where $a, b \in \mathbb{Z}$.

22-(MAT/12_HL_Summer_2018_Q7)-*ComplexNumbers*

Consider the distinct complex numbers $z = a + \mathrm{i}b$, $w = c + \mathrm{i}d$, where $a, b, c, d \in \mathbb{R}$.

(a) Find the real part of $\dfrac{z+w}{z-w}$. [4]

(b) Find the value of the real part of $\dfrac{z+w}{z-w}$ when $|z| = |w|$. [2]

23–(MAT/11_HL_Summer_2018_Q11)–*ComplexNumbers*

Consider $w = 2\left(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\right)$.

(a) (i) Express w^2 and w^3 in modulus-argument form.

(ii) Sketch on an Argand diagram the points represented by w^0, w^1, w^2 and w^3. [5]

These four points form the vertices of a quadrilateral, Q.

(b) Show that the area of the quadrilateral Q is $\frac{21\sqrt{3}}{2}$. [3]

Let $z = 2\left(\cos\frac{\pi}{n} + \mathrm{i}\sin\frac{\pi}{n}\right)$, $n \in \mathbb{Z}^+$. The points represented on an Argand diagram by $z^0, z^1, z^2, \ldots, z^n$ form the vertices of a polygon P_n.

(c) Show that the area of the polygon P_n can be expressed in the form $a\left(b^n - 1\right)\sin\frac{\pi}{n}$, where $a, b \in \mathbb{R}$. [6]

24-(MAT/10_HL_Winter_2018_Q8)-*ComplexNumbers*

[Maximum mark: 7]

Consider the equation $z^4 + az^3 + bz^2 + cz + d = 0$, where $a, b, c, d \in \mathbb{R}$ and $z \in \mathbb{C}$.
Two of the roots of the equation are $\log_2 6$ and $\mathrm{i}\sqrt{3}$ and the sum of all the roots is $3 + \log_2 3$.
Show that $6a + d + 12 = 0$.

25-(MAT/10_HL_Winter_2018_Q11)-*ComplexNumbers*

(a) Find the roots of $z^{24} = 1$ which satisfy the condition $0 < \arg(z) < \frac{\pi}{2}$, expressing your answers in the form $re^{i\theta}$, where $r, \theta \in \mathbb{R}^+$. [5]

(b) Let S be the sum of the roots found in part (a).

(i) Show that $\operatorname{Re} S = \operatorname{Im} S$.

(ii) By writing $\frac{\pi}{12}$ as $\left(\frac{\pi}{4} - \frac{\pi}{6}\right)$, find the value of $\cos\frac{\pi}{12}$ in the form $\frac{\sqrt{a} + \sqrt{b}}{c}$, where a, b and c are integers to be determined.

(iii) Hence, or otherwise, show that $S = \frac{1}{2}\left(1 + \sqrt{2}\right)\left(1 + \sqrt{3}\right)(1 + i)$. [11]

26-(MAT/10_HL_Winter_2019_Q5)-*ComplexNumbers*

[Maximum mark: 7]

Consider the equation $z^4 = -4$, where $z \in \mathbb{C}$.

(a) Solve the equation, giving the solutions in the form $a + ib$, where $a, b \in \mathbb{R}$. [5]

(b) The solutions form the vertices of a polygon in the complex plane. Find the area of the polygon. [2]

27-(MAT/10_HL_Winter_2020_Q4)-*ComplexNumbers*

[Maximum mark: 5]

Consider the equation $\dfrac{2z}{3-z^*} = \mathrm{i}$, where $z = x + \mathrm{i}y$ and x, $y \in \mathbb{R}$.

Find the value of x and the value of y.

28-(MAT/10_HL_Winter_2020_Q7)-*ComplexNumbers*

Consider the complex numbers $z_1 = \cos\frac{11\pi}{12} + \mathrm{i}\sin\frac{11\pi}{12}$ and $z_2 = \cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}$.

(a) (i) Find $\frac{z_1}{z_2}$

(ii) Find $\frac{z_2}{z_1}$ [3]

(b) 0, $\frac{z_1}{z_2}$ and $\frac{z_2}{z_1}$ are represented by three points O, A and B respectively on an Argand diagram. Determine the area of the triangle OAB. [2]

29-(MathAA/11_HL_Summer_2021_Q7)-*ComplexNumbers*

[Maximum mark: 8]

Consider the quartic equation $z^4+4z^3+8z^2+80z-400=0$, $z\in\mathbb{C}$.

Two of the roots of this equation are $a+b\mathrm{i}$ and $b+a\mathrm{i}$, where $a, b\in\mathbb{Z}$.

Find the possible values of a.

30–(MathAA/10_HL_Winter_2021_Q12)–*ComplexNumbers*

[Maximum mark: 22]

Consider the equation $(z-1)^3 = \mathrm{i}$, $z \in \mathbb{C}$. The roots of this equation are ω_1, ω_2 and ω_3, where $\mathrm{Im}(\omega_2) > 0$ and $\mathrm{Im}(\omega_3) < 0$.

(a) (i) Verify that $\omega_1 = 1 + \mathrm{e}^{\mathrm{i}\frac{\pi}{6}}$ is a root of this equation.

(ii) Find ω_2 and ω_3, expressing these in the form $a + \mathrm{e}^{\mathrm{i}\theta}$, where $a \in \mathbb{R}$ and $\theta > 0$. [6]

The roots ω_1, ω_2 and ω_3 are represented by the points A, B and C respectively on an Argand diagram.

(b) Plot the points A, B and C on an Argand diagram. [4]

(c) Find AC. [3]

Consider the equation $(z-1)^3 = \mathrm{i}z^3$, $z \in \mathbb{C}$.

(d) By using de Moivre's theorem, show that $\alpha = \dfrac{1}{1 - \mathrm{e}^{\mathrm{i}\frac{\pi}{6}}}$ is a root of this equation. [3]

(e) Determine the value of $\mathrm{Re}(\alpha)$. [6]

31-(MathAA/11_HL_Summer_2022_Q9)-*ComplexNumbers*

[Maximum mark: 6]

Consider the complex numbers $z_1 = 1 + b\mathrm{i}$ and $z_2 = (1 - b^2) - 2b\mathrm{i}$, where $b \in \mathbb{R}$, $b \neq 0$.

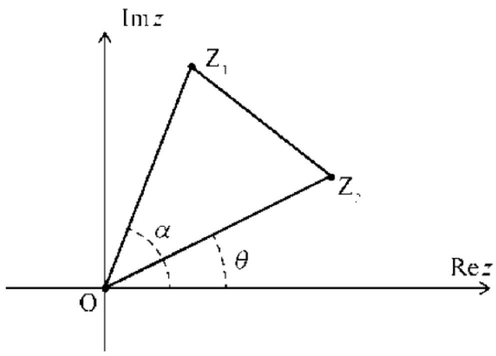
(a) Find an expression for $z_1 z_2$ in terms of b. [3]

(b) Hence, given that $\arg(z_1 z_2) = \dfrac{\pi}{4}$, find the value of b. [3]

32-(MathAA/12_HL_Summer_2022_Q12)**-***ComplexNumbers*

[Maximum mark: 18]

In the following Argand diagram, the points Z_1, O and Z_2 are the vertices of triangle Z_1OZ_2 described anticlockwise.



The point Z_1 represents the complex number $z_1 = r_1 e^{i\alpha}$, where $r_1 > 0$. The point Z_2 represents the complex number $z_2 = r_2 e^{i\theta}$, where $r_2 > 0$.

Angles α, θ are measured anticlockwise from the positive direction of the real axis such that $0 \le \alpha, \theta < 2\pi$ and $0 < \alpha - \theta < \pi$.

(a) Show that $z_1 z_2{}^* = r_1 r_2 e^{i(\alpha - \theta)}$ where $z_2{}^*$ is the complex conjugate of z_2. [2]

(b) Given that $\operatorname{Re}(z_1 z_2{}^*) = 0$, show that Z_1OZ_2 is a right-angled triangle. [2]

In parts (c), (d) and (e), consider the case where Z_1OZ_2 is an equilateral triangle.

(c) (i) Express z_1 in terms of z_2.

(ii) Hence show that $z_1{}^2 + z_2{}^2 = z_1 z_2$. [6]

Let z_1 and z_2 be the distinct roots of the equation $z^2 + az + b = 0$ where $z \in \mathbb{C}$ and $a, b \in \mathbb{R}$.

(d) Use the result from part (c)(ii) to show that $a^2 - 3b = 0$. [5]

Consider the equation $z^2 + az + 12 = 0$, where $z \in \mathbb{C}$ and $a \in \mathbb{R}$.

(e) Given that $0 < \alpha - \theta < \pi$, deduce that only one equilateral triangle Z_1OZ_2 can be formed from the point O and the roots of this equation. [3]

33–(MathAA/10_HL_Winter_2022_Q5)–*ComplexNumbers*

[Maximum mark: 7]

Consider the equation $z^4 + pz^3 + 54z^2 - 108z + 80 = 0$ where $z \in \mathbb{C}$ and $p \in \mathbb{R}$.

Three of the roots of the equation are $3 + \mathrm{i}$, α and α^2, where $\alpha \in \mathbb{R}$.

(a) By considering the product of all the roots of the equation, find the value of α. [4]

(b) Find the value of p. [3]